

Combination of laser keyhole and conduction welding: Dissimilar laser welding of niobium and Ti-6Al-4V



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ABSTRACT

Pulsed Nd:YAG laser welding of pure niobium plate to titanium alloy Ti-6Al-4V sheet in butt joint is studied regarding the laser/metal interaction modes. To obtain the optimized process parameters in dissimilar welding of Ti-6Al-4V/Nb, the melting ratio of laser beam energy for each weld counterpart is evaluated experimentally. Different laser welding modes of keyhole and conduction are predicted regarding the absorbed energy from the similar laser pulses on each weld counterpart. Laser keyhole and conduction welding were observed simultaneously through direct visualization of laser interaction with dissimilar metals using High Speed Imaging (HSI) system.

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1. Introduction

Laser welding of dissimilar materials is used for several years and has been established as one of the most efficient methods in joining of many kinds of materials [1]. Interaction of laser beam with the weld counterparts in dissimilar welding may be unlike and different materials interact through conduction or keyhole welding under laser radiations [2]. Generally, it is accepted that coupling of the laser beam with metal occurs either through the melt pool surface-conduction welding- or a deep vapor filled cavity known as keyhole [3]. The amount of laser beam energy absorbed by the target is mainly governed by conduction/keyhole interactions between the laser beam and workpieces [4]. In conduction laser welding of metals, typically less than 20% of the laser beam energy is used efficiently for melting while in laser keyhole welding this portion increases up to 70–90% [5]. Regarding the energy absorption factor, optimization of the laser welding mode is still of interest for many researchers [6–8] especially in dissimilar welding [9]. Assuncao et al. studied the behavior of different metals in laser welding while transition from conduction to keyhole modes [6]. Their experiments showed that material thermal properties (thermal conductivity, melting and vaporization temperatures and specific heat) have the most important role in transition between the laser welding modes. Sibillano et al. developed a real-time monitoring technique based on the analysis of the plasma plume

optical spectra generated during laser welding, to determine the laser welding mode [10].

Among dissimilar welding often it is observed that laser beam is effectively paired with one of the weld counterparts while its coupling with the other side is very weak [11]. Thus the melt pool size on one of the weld sides is increased more than the other one. The melt pool at dissimilar interface is pooled toward the side with higher laser absorption [12,13]. Chakraborty showed that because of differences in thermo-physical properties of Cu and Ni, in dissimilar Nd:YAG laser welding, the joint melt pool is mostly developed in the nickel side [11]. Deflection of the melt pool toward the nickel side was predicted through numerical modeling and confirmed with experimental observations [11]. In dissimilar laser welding of stainless steel and nickel, Hu et al. reported that because of the higher thermal conductivity of Ni, the melt pool is almost leaning toward the steel side and there are no hint or reference to the possibility of simultaneous different laser welding modes [12].

Ti-6Al-4V is the most widely used titanium alloy in aerospace industry because of its high level of strength-to-weight ratio and niobium is a refractory metal with highest critical temperature among metals that becomes superconductor at 9 K (which is just above the liquid helium temperature (4 K)). Dissimilar welding of titanium alloy Ti-6Al-4V to niobium is used in high performance structures like particle accelerators [13]. Although dissimilar welding of Ti-6Al-4V and niobium is proven to be a challenge due to very different thermo-physical properties of the joint counterparts but welding of these two materials using a pulsed Nd:YAG laser has been reported recently [14].

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However, different responds of the Ti alloy and Nb plates to laser beam, asymmetric development of the weld pool at interface, and adjusting laser beam center respect to the plates interface, although were recognized to be very important in the above mentioned welding process, were not considered properly yet. Here in this study the aforementioned issues are deeply investigated comparing separate

metals interactions with the laser beam and direct observations of welding processes by use of High Speed Imaging.

In the present study, first a short description of the background experimental work is presented, then it will be focused on laser-metal interactions and energy absorption modes from pulsed Nd:YAG laser beam by Ti-6Al-4V and Nb plates. For similar pulsed laser parameters, melting ratio of single pulses is evaluated for both of the weld counterparts. Laser welding modes (conduction or keyhole) can be estimated regarding the melting ratio values evaluated for Ti-6Al-4V and Nb. Moreover, High Speed Imaging (HSI) technique is used to directly observe laser interactions with titanium alloy and niobium plates individually and in actual dissimilar welding which its outcomes can be used to confirm the laser welding modes predicted using single pulse melting experiments of Ti-6Al-4V and Nb.

2. Experimental setup

Two different Nd:YAG laser sources were used for pulsed laser welding in this study. In the first series of experiments (single pulse tests) a pulsed Nd:YAG laser (model IQL-10) with a maximum mean laser power of 400 W was used as laser source. Square shape pulses are the standard output of this laser. The available ranges for the laser parameters are 1 to 1000 Hz for pulse frequency, 0.2 to 20 ms for pulse duration, and 0 to 40 J for pulse energy. Since the average laser power

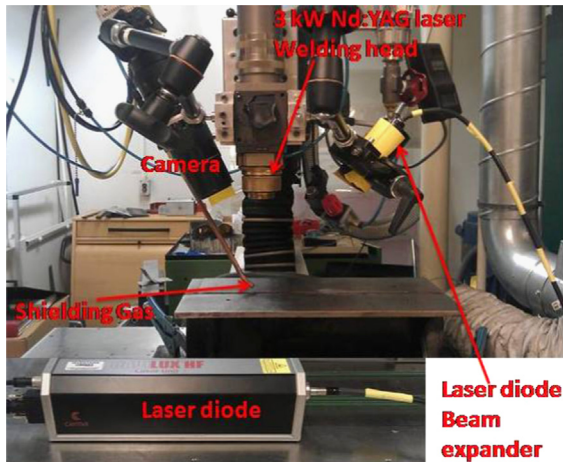


Fig. 1. Laser welding setup with High Speed Imaging system.

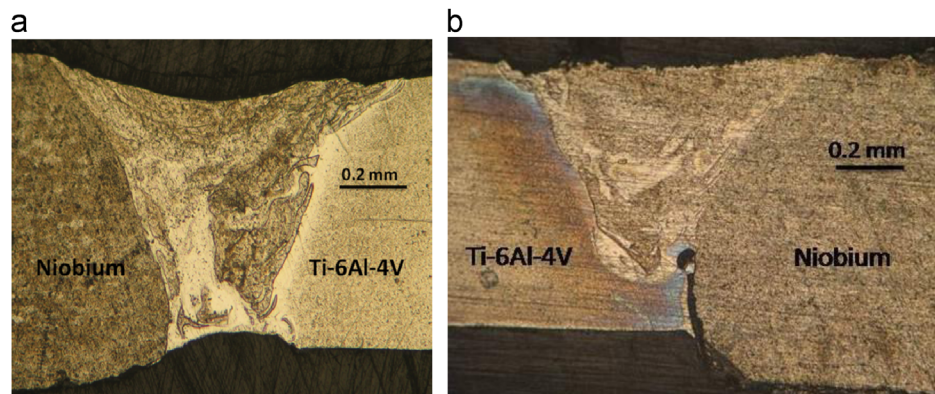


Fig. 2. Cross sections of two Ti-6Al-4V/Nb laser welds obtained with similar laser pulse energy (9 J) and duration (6 ms), (a) complete interface joining with adjustment of laser spot center relative to the plates interface and (b) deflected melt pool with partial interface joining.

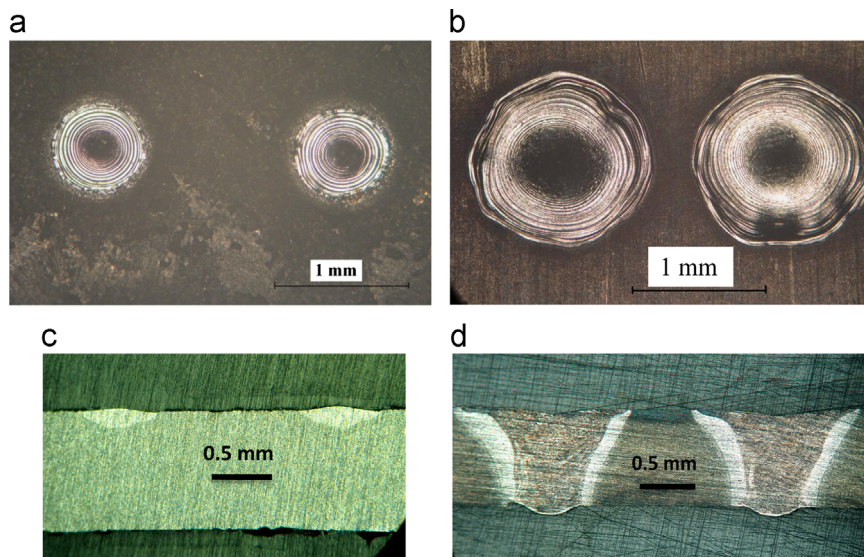


Fig. 3. Images of weld spots obtained with laser pulses 9 J in 6 ms. (a) Niobium surface, (b) Ti-6Al-4V surface, (c) Niobium cross section, and (d) Ti-6Al-4V cross section.

could not be more than 400 W, it is evident that no arbitrary combination of pulse energy and pulse frequency could be used. The focusing length of laser beam is 75 mm with 400 μm minimum spot size. Details of the laser welding experimental set-up are represented in previous report [4].

Another Nd:YAG laser source, model Haas HL3006D, with focusing optics of 200 mm, and about $\sim 600 \mu\text{m}$ spot size, was used for second series of dissimilar welding experiments accompanying with High Speed Imaging (HSI). This laser is able to radiate in continuous wave and pulsed mode (this mode was actually used in the experiments) with up to 3 kW laser power and predetermined parameters. All welding processes with Haas laser were recorded using a High Speed Imaging system. A high speed camera (model Photron SA1 high-speed camera with a Micro-Nikor 105-mm lens) was operated at 4500 and 6500 frames per second for imaging the laser/metal interaction area that was illuminated with a correspondingly pulsed Cavitax HF diode laser system. The fiber-guided pulsed diode laser (with 500 W peak power) is operated at 808 nm wavelength to illuminate the workpiece surface. The camera had a narrow band-pass filter for the illumination laser wavelength (808 nm) and preventing other light spectra from Nd: YAG laser and plasma plume to enter and disturb the imaging processes. The camera was positioned laterally, at an angle of 55° to the surface (Fig. 1). Beside qualitative analysis [15], quantitative properties can be measured from the images obtained by High Speed Imaging technique [16]. Materials used for welding were pure niobium plates with 1 mm thickness and 0.85 mm thickness Ti-6Al-4V plates. Because of sensitivity of both metals, shielding gas on top and bottom of the laser interacting area were required. Pure argon gas was used as shielding gas commonly on top and bottom of the weld through a particular design [14].

3. Results and discussions

3.1. Background work

In the first series of experiments using the 400 W pulsed Nd:YAG laser, at certain process parameters sound welds between Ti-6Al-4V and niobium was obtained, see Fig. 2(a). The complexity in laser welding of this couple of metals was mainly found in relation to the different thermo-physical properties of Ti-6Al-4V and niobium, that results to asymmetric development of the weld pool formed under laser radiation (Fig. 2(b)). Laser beam is absorbed more efficiently by titanium alloy while its coupling with the niobium is weaker. Moreover, energy received by the Nb side is dissipated to the bulk metal very fast whereas Ti-6Al-4V sustains energy for longer times because of its lower thermal conductivity [14].

Table 1

Laser welding process parameters in niobium and Ti-6Al-4V radiation with single laser pulses.

Pulse energy (J)	Pulse duration (ms)	Laser repetition rate (Hz)	Pulse peak power (kW)	Process speed (mm/s)
9	6	20	1.5	35

Table 2

Thermo-physical properties of Niobium and Ti-6Al-4V [18,19].

Properties Material	Melting point (°C)	Boiling point (°C)	Density* (g/cm ³)	Specific heat *C _p [J/(g K)]	Thermal conductivity* [W/(m K)]	Coefficient of linear thermal expansion* [m/(m K)]	Latent heat of fusion (J/g) L _f
Niobium	2468	4900	8.55	0.27	53.7	7.1×10^{-6}	285
Ti-6Al-4V	1655	3315	4.43	0.61	6.7	8.6×10^{-6}	290

* Measured at 20 °C.

In order to achieve an acceptable weld with complete joint penetration along the two metal edges, the beam had to be displaced towards the titanium side. However, the question remained about the physical phenomena governing such behavior and the solution. Thus, further observations and analysis about the interaction of laser beam with Ti-6Al-4V and niobium plates were required. So, two sets of follow up experiments were designed including interaction of single laser pulses with each metal and High Speed Imaging of dissimilar laser welding process.

3.2. Single laser pulse interactions with Ti-6Al-4V and niobium

To study the laser pulse interaction with the weld counterparts, the Ti-6Al-4V and niobium plates were separately radiated with a chain of laser pulses such that successive pulses did not overlap. In this case the energy absorption from single laser pulses with niobium and titanium alloy and responses of the plates to the absorbed energy can be estimated directly. Images of the top surfaces and cross sections of the two consecutive laser spots on niobium and Ti-6Al-4V plates are presented in Fig. 3a–d. The process parameters used in this experiment are listed in Table 1. The laser spot diameter throughout the experiments was kept constant at 0.4 mm.

The average diameter of the resulting weld spot widths on Ti-6Al-4V plate was $W_{Ti}=1.42 \text{ mm}$ with the average melt volume of $V_{Ti-6Al-4V}=0.367 \text{ mm}^3$ while on niobium plate the average diameter was $W_{Nb}=0.68 \text{ mm}$ and respected volume is $V_{Nb}=0.031 \text{ mm}^3$, i.e. the niobium melted volume is about 12 times smaller than Ti-6Al-4V melted volume.

To estimate the amounts of energy consumed to melt the plates (in response to single laser pulses) a suitable welding variable known as *melting ratio* can be used [17]. The melting ratio (η_{mr}) for a single laser spot welding is defined as the ratio of energy required to melt the weld metal to the energy incident on the workpiece [17]:

$$\eta_{mr} = V \cdot \rho \cdot \left(\int_{T_0}^{T_m} C_p \cdot dT + \Delta H_m \right) / E_p \quad (1)$$

Where V is the melt volume of the spot-weld pool, " ρ " the density, C_p the heat capacity, T the temperature, T_0 the initial temperature (room temperature), " T_m " the melting temperature, " H_m " the latent heat of fusion of the material, and E_p is the energy of the incident laser pulse. To estimate the portion of the laser pulse energy consumed to melt the target plates, Eq. (1) can be simplified by using a constant value for the C_p of Ti-6Al-4V and Nb as indicated in Table 2. A simple relation

Table 3

Comparison of melting ratio in single laser pulses interacting with Ti-6Al-4V and Nb.

Plate material	Volume of the single pulse melt pool (mm ³)	Mass of the single pulse melt pool (g)	Required energy for melting (J)	Melting ratio (η_m)
Ti-6Al-4V	0.367	1.61×10^{-3}	2.07	0.23
Nb	0.031	0.265×10^{-3}	0.26	0.029

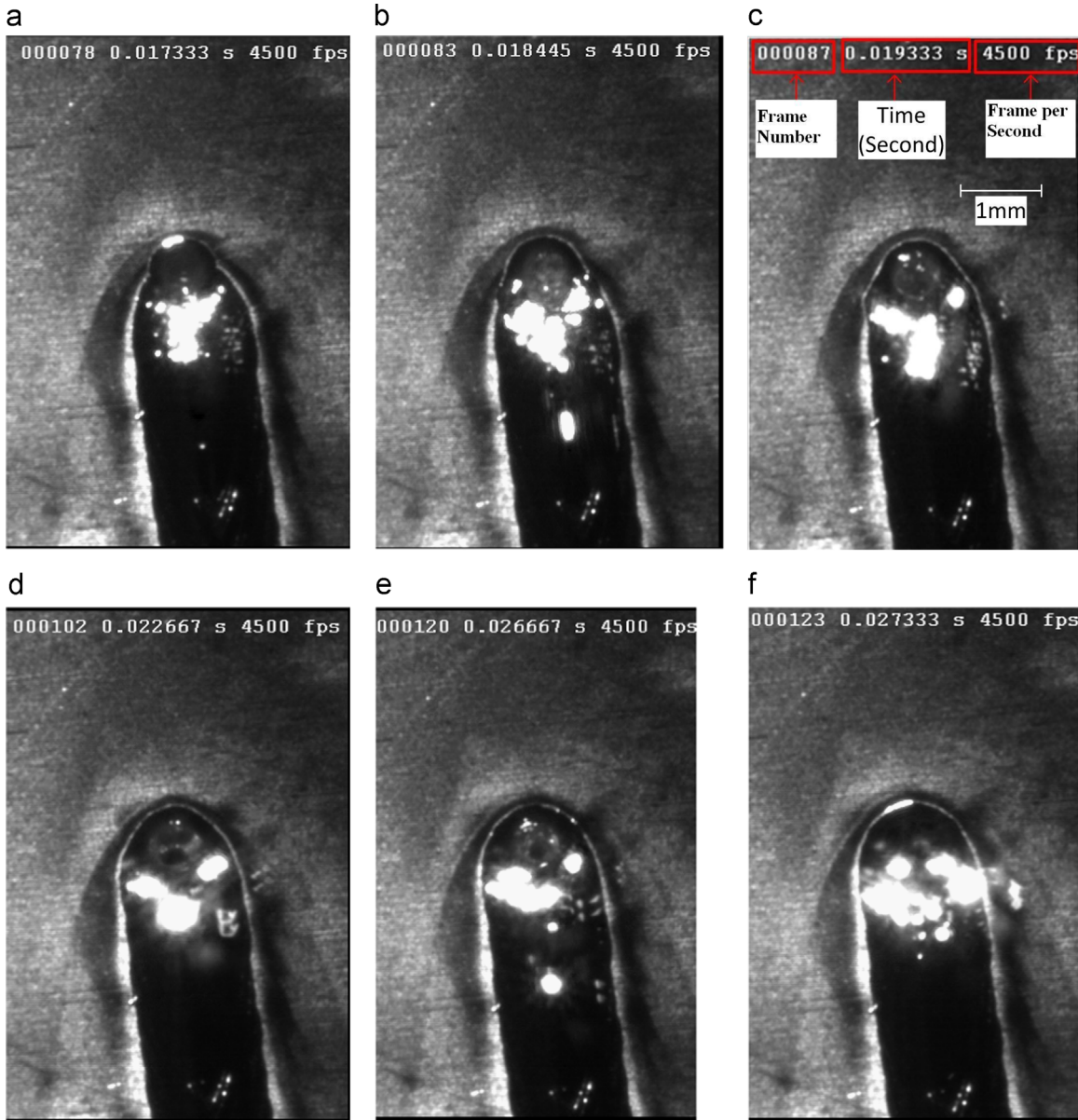


Fig. 4. Bead on plate welding of Ti-6Al-4V with a single laser pulse is recorded using HSI, part (a) shows laser pulse beginning, (b) represents keyhole initiation, (c) keyhole completed, (d) keyhole receives back side of the plate, (e) pulse ramping down, and (f) end of laser pulse.

can be used for calculating the required energy to heat up and melting “ Q ” as:

$$Q = mC\Delta\theta + mL_f = m \times (C\Delta\theta + L_f) \quad (2)$$

Where, C_p refers to the specific heat, ΔT represents the temperature rise from room temperature to the melting point and L_f is the latent heat of fusion. The parameter “ m ” here refers to the mass of the melted material.

It is clear that the possibility of keyhole formation and evaporation inside the melt pool is not considered in the above calculations. Only heating up to the fusion point and melting was calculated in the Eq. (2). It can be anticipated that in the case of evaporation (formation of keyhole) the amount of the absorbed energy would be more than that estimated by Eq. (2).

Using the densities given in Table 2, mass of the melted niobium and titanium in single pulse welding can be calculated as: $m_{\text{Ti-6Al-4V}} = 1.61 \times 10^{-3}$ gr and $m_{\text{Nb}} = 0.265 \times 10^{-3}$ g.

By replacing these masses in Eq. (2) for each metal, following results for the consumed energy will be obtained: $Q_{\text{Ti-6Al-4V}} = 2.07$ J and $Q_{\text{Nb}} = 0.26$ J.

Putting these quantities in Eq. (1) the melting ratios for each kind of metals can be calculated as:

$$\eta_{mr(\text{Ti-6Al-4V})} = Q_{\text{Ti-6Al-4V}}/E_p = 0.23; \quad \eta_{mr(\text{Nb})} = Q_{\text{Nb}}/E_p = 0.029$$

The melt pool volumes, masses, required energy for melting, and melting ratio values for interaction of separate laser pulses with titanium alloy and niobium are briefly presented in Table 3.

It means that from 9 J energy of the laser pulse 2.07 J is consumed to melt the titanium alloy while only 0.26 J is used for niobium melting. It is known that energy absorption from laser pulse continues after melting to heating the melt up to the vaporization temperature and keyhole formation. Keyhole creation results to significant increase in the energy absorption through the inverse Bremsstrahlung mechanism and Fresnel reflections [4].

Obviously whole amount of the absorbed energy by metal plate is not considered in the melting ratio because of the plate heat

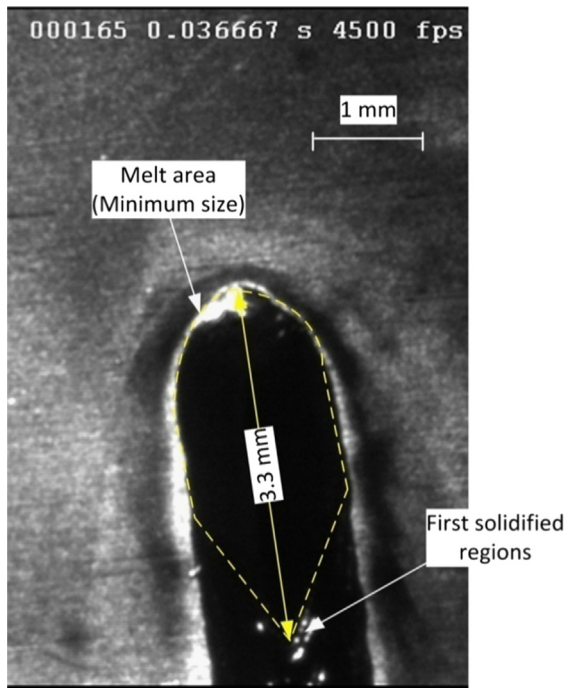


Fig. 5. A typical frame from the video of the Ti-6Al-4V weld capillary. This frame shows the minimum size of liquid phase melt pool at end of the off time of pulse period and before beginning of the next pulse.

losses through conduction, convection and radiation mechanisms. As presented in Table 3 the melting ratio of Ti-6Al-4V for single laser pulses is about one order of magnitude greater than that for Nb plate, i.e. laser metal interaction and energy absorption mechanisms were significantly different. Thus, it can be anticipated that the laser pulse had interacted with niobium in conduction mode while it is coupled with Ti-6Al-4V plate in keyhole mode. It is known that in conduction laser welding energy is absorbed by workpiece surface, whereas when keyhole forms, the incoming laser beam is trapped inside the keyhole and mostly absorbed by successive reflections from keyhole walls which are recognized as Fresnel reflections. Resultantly in the first case since laser beam energy is only absorbed through surface a large portion of beam energy is lost by surface reflections, while in keyhole mode, more energy is absorbed by workpiece because of the beam trapping phenomenon in the laser created keyhole. It is found that up to 93% of laser beam energy can be absorbed by metal workpiece in laser keyhole welding of aluminum [5]. To verify the proposition made in the experiments involving single pulse interactions with Ti-6Al-4V and niobium plates, a High Speed Imaging system was used to observe the process directly.

3.3. High Speed Imaging of pulsed laser dissimilar welding

The use of high speed cameras in laser welding research dates back to 1985 when Arata and co-workers used a camera with 6000 frames per second (fps) to observe the process [20]. Progress of high speed digital cameras has made the HSI easier to use, and currently

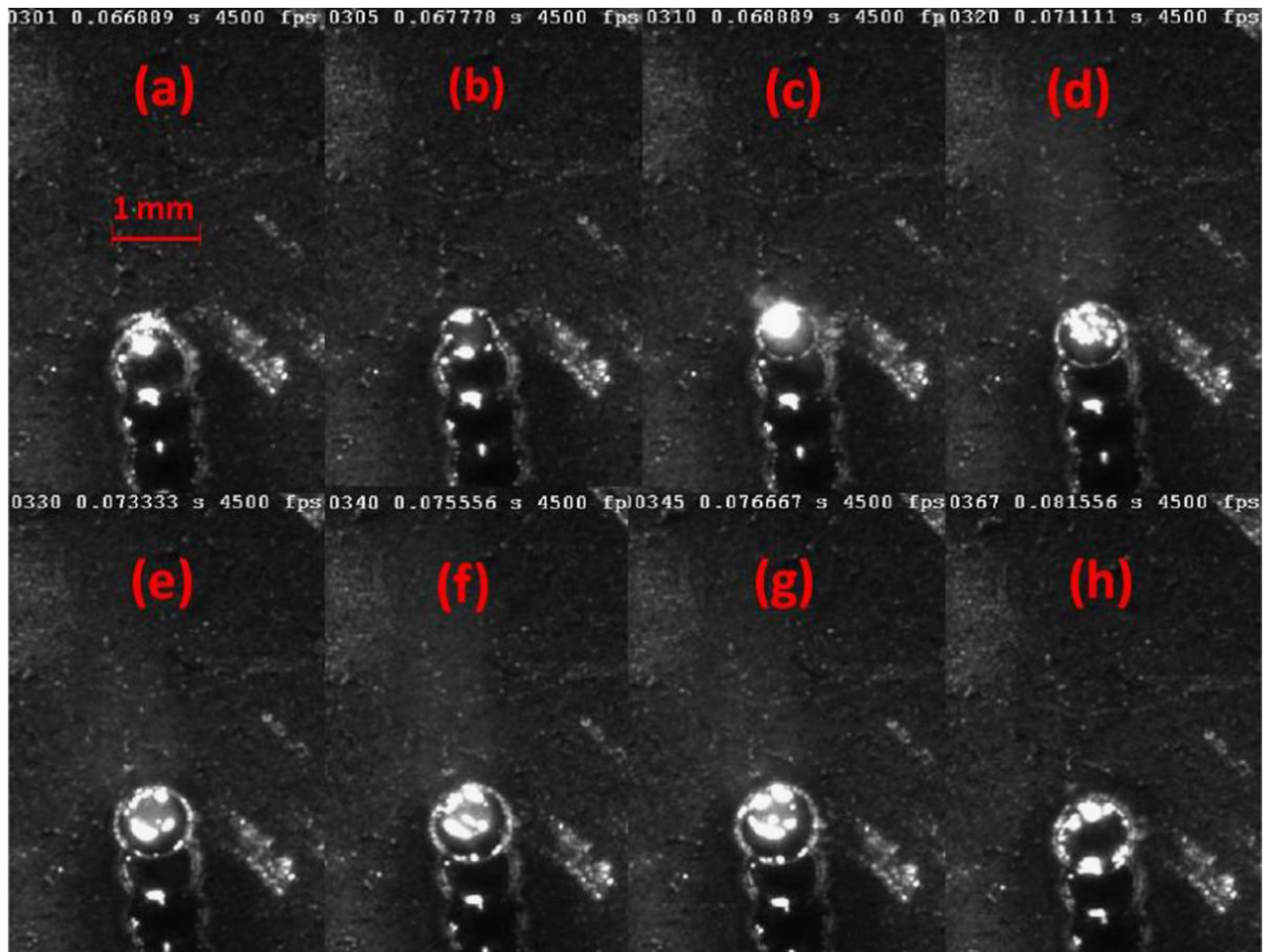


Fig. 6. High Speed imaging of a laser pulse interaction with niobium plate, (a) shows the first signs of the pulse strike, (b–f) typical frames showing development of the weld pool in conduction mode, (g) end of laser pulse on time interval (10 ms) with maximum melt pool size, and (h) complete solidification of the melt pool.

high-speed photography with up to 20,000 fps has become standard instrument in many research groups [15]. High Speed Imaging method is used in this research to directly observe and record pulsed laser interaction with Nb and Ti-6Al-4V plates separately and their combination in dissimilar laser welding. Our investigations here is concentrated to distinguish probably different laser welding modes, to confirm the results obtained in pervious section.

At first Ti-6Al-4V and then niobium plates were radiated by predetermined pulsed mode of a high power HAAS Nd:YAG laser. The laser pulse energy and duration were fixed at 15 J and 10 ms respectively with 50 Hz pulse repetition frequency. The laser beam radiates on a fixed point after passing through the concentration system and workpiece moves with 25 mm/s speed relatively. All laser metal interaction processes were directly observed and recorded with a high speed camera (working at 4500 frames per second) accompanied with laser diode illumination as explained in experimental setup section and Fig. 1 with more details.

Typical interactions of a single laser pulse period (having 10 ms on and 10 ms off durations) with the Ti-6Al-4V plate is recorded using 90 frames of HSI, from which 6 distinct frames of the pulse on time half period are selected and shown in Fig. 4. According to Fig. 4 (a)–(f) interaction of the laser pulse with metal plate begins at 0.017333 s (frame number 78) and finishes at 0.027333 s (frame number 123). Fig. 4(b) shows the first signs of the boiling and evaporation on top of the liquid metal just in laser interaction zone. After less than 2 ms the keyhole is completely formed as Fig. 4 (c) represents. Keyhole penetrates the plate thickness (0.85 mm) and after a short time reaches the back side of the plate (Fig. 4(d)).

As shown in Fig. 4(e) near the pulse ending time, because of the gradual decrease in laser power density, keyhole size becomes smaller. After 10 ms, the on time half period of pulse finishes at $t=0.027333$ s and keyhole collapses rapidly (Fig. 4(f)). After that there would be a cooling interval equal to 10 ms (off time, half period of the pulse) before the next pulse. Fig. 5 shows the minimum length (3.3 mm) of the liquid capillary of melt pool just before the next pulse start. It should be noted that for the present welding parameters of Ti-6Al-4V plate, the melt does not solidify completely between pulses. It means that after passing the off time, half period of laser pulse, the metal under laser radiation is in the liquid form and the next pulse strikes to the molten state of the workpiece.

A complete sequence of laser pulse interaction with niobium plate, from pulse initiation followed by melt pool development and solidification are represented through 8 distinct HSI frames in Fig. 6. As shown in Fig. 6(a), interaction of the laser pulse with niobium begins at 0.066 s (frame number 301) and finishes at 0.076 s (frame number 345). A circular melt pool (not elliptical shape similar to Ti-6Al-4V weld pool) is formed just around the laser beam spot on Nb plate surface through conduction melting. The maximum diameter of this shallow circular melt pool is achieved at end of the pulse on time interval (0.076 s) and is around 0.7 mm (Fig. 3(c)). After pulse on time half period, laser stops radiation and circular melt area lose out its heat to become solidified. As shown in Fig. 6(h) about 5 ms after laser stops radiation, the melt pool is completely solidified. It means that niobium behavior under laser radiation is quite different to that of

Table 4
results summery of High Speed Imaging of interaction of a single laser pulse (10 ms on – 10 ms Off) with Ti-6Al-4V and Nb plates separately.

Material	Laser pulse beginning	Keyhole formation time	Melt pool sizes at end of pulse on time (at 10 ms)	Solidification completed time	melt pool sizes at end of laser pulse period (at 20 ms)
Ti-6Al-4V	0 ms	< 1 ms	3.9 mm × 1.4 mm	Never	3.3 mm × 1.4 mm
Niobium	0 ms	Never	A circle with 0.7 mm diameter	≈ 15 ms	0

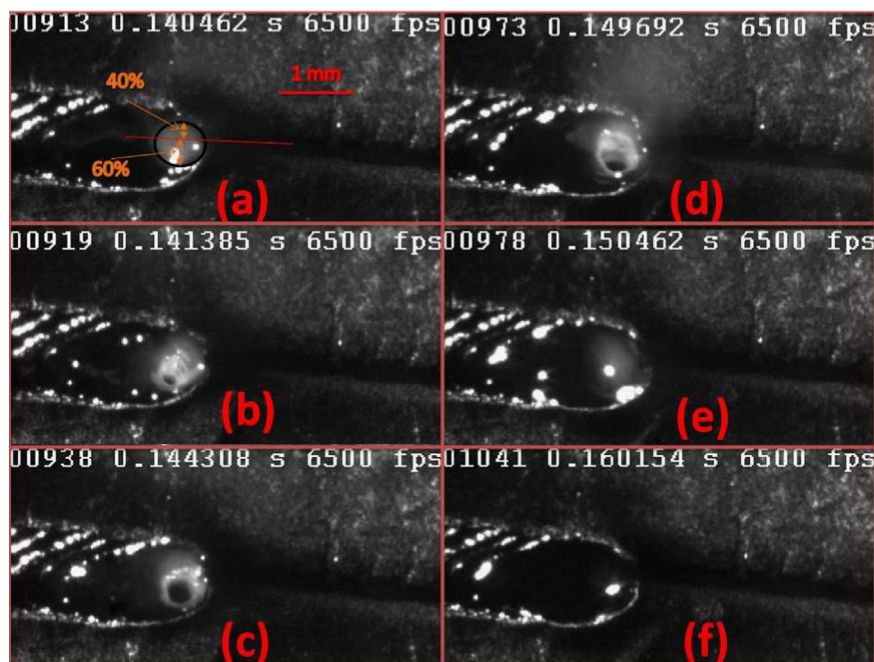


Fig. 7. Chain of 6 High Speed Imaging frames of a single 20 ms laser pulse interaction with dissimilar Ti-6Al-4V/Nb interface. Part (a) shows the beginning of laser pulse, (b) after ~1 ms the first signs of keyhole on Ti-6Al-4V is appeared, (c) after ~4 ms keyhole is completely developed on Ti-6Al-4V, (d) after ~9.5 ms keyhole is still on Ti-6Al-4V with no shift to the Nb side, (e) end of laser pulse with maximum melt pool sizes, and (f) after passing the pulse off time (~20 ms) the liquid flow can be seen in the pool, it means that complete solidification of the melt pool doesn't occur between successive pulses.

Ti-6Al-4V that remains in liquid state between successive pulses. It is because of very different thermo-physical properties of Ti-6Al-4V and niobium as presented in Table 2. Niobium absorbs less energy from laser beam with similar pulse parameters than Ti-6Al-4V but dissipates its energy about 10 times faster than Ti-6Al-4V. The subsequent stages of laser pulse interactions with titanium alloy and niobium is briefly presented in Table 4.

In the next step, dissimilar welding of Ti-6Al-4V and niobium was performed such that the laser beam center was mostly positioned on Ti-6Al-4V side (60% beam diameter is located on titanium alloy and 40% on niobium) to obtain a sound weld with acceptable tensile strength [14].

High Speed Imaging of a complete period of a single laser pulse action (equal to 20 ms, 10 ms on and 10 ms off intervals) in dissimilar welding of Ti-6Al-4V/Nb is represented in Fig. 7. In fact during 20 ms period of a single laser pulse, about 130 HSI frames have been recorded, that 6 of them were selected and are shown in Fig. 7. Fig. 7(a) shows that laser pulse begins to radiate at the interface of two plates such that about 60% laser spot diameter is located on Ti-6Al-4V and the remained part on niobium. It is seen that at initiation of new laser pulse, the earlier pulse melt pool is still remained in liquid state. A short time after pulse initiation (about 0.92 ms), first signs of the keyhole appears at the Ti alloy side as shown in Fig. 7(b). In the remained 9 ms on time of the laser pulse, the melt pool develops at the Ti-6Al-4V/ Nb interface but with conduction mode on the Nb and keyhole mode on the other one. Two typical frames of this time interval are shown in Figs. 7(c) and (d). (e) shows the final frame of the pulse on time episode. After that, laser stops radiation for 10 ms, such that solidification of the melted materials is the most governing process. But the melt pool does not solidify entirely till opening of the next pulse as shown by Fig. 7(f). It is clearly seen that while laser radiates on the dissimilar interface no signs of evaporation is observable near the niobium edge whereas a clear evaporated hole appears on the Ti-6Al-4V side. In the other words the laser beam spot fraction located on the Ti-6Al-4V side, creates a keyhole while it couples with the niobium only through the melt surface, which is known as conduction welding.

4. Conclusions

In dissimilar welding of Ti-6Al-4V and niobium plates by pulsed Nd:YAG laser interactions of separate and mutual metal plates with laser beam were studied using a High Speed Imaging system. Melting ratio of single spot welds on Ti-6Al-4V and Nb plates by distinct laser pulses were evaluated and compared. Following conclusions can be obtained from this research:

1. Melting ratio of separate laser pulses with similar parameters on Ti-6Al-4V is one order of magnitude greater than melting ratio of Nb plate.
2. In pulsed Nd:YAG laser welding of Ti-6Al-4V/Nb keyhole forms about 0.92 ms after pulse initiation on titanium alloy side.
3. Under pulsed laser radiation with the same process parameters, a circular melt pool forms on niobium surface while in the case of Ti-6Al-4V the weld pool has elliptical appearance.
4. The melt pool formed on Ti-6Al-4V does not freeze between succeeding pulses but in case of Nb it solidifies. However, in

welding the two metals together, the common weld pool remained in liquid state between pulses.

5. In dissimilar welding of Ti-6Al-4V/Nb with pulsed Nd:YAG laser, the both known laser welding modes, i.e conduction and key-hole welding can occur simultaneously.

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